

### 16.8.1 Problems

16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is  $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$ , where  $\Lambda(\bullet)$  is the logistic cdf in equation (16.7). The logit model parameter estimates and their standard errors are

$$\begin{matrix} \hat{\gamma}_1 + \hat{\gamma}_2 DTIME = & -0.2376 + 0.5311 DTIME \\ (se) & (0.7505) \quad (0.2064) \end{matrix}$$

- Calculate the estimated probability that a person will choose automobile transportation given that  $DTIME = 1$ .
- Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that  $DTIME = 1$ . How does this result compare to the logit estimate? [Hint: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]
- Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?
- Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

$$\text{Logistic cdf: } F(x) = \frac{e^x}{1+e^x}$$

$$\frac{dF(x)}{dx} = \frac{e^x(1+e^x) - e^x \cdot e^x}{(1+e^x)^2} = \frac{e^x}{1+e^x} \cdot \frac{1}{1+e^x} = h \cdot (1-h), \text{ where } h = \frac{e^x}{1+e^x} \quad \dots (*)$$

$$a. P(AUTO = 1 | DTIME = 1) = \Lambda(-0.2376 + 0.5311 \times 1) = \Lambda(0.2935) = \frac{e^{0.2935}}{1+e^{0.2935}} = 0.5728$$

$$b. P(AUTO = 1 | DTIME = 1) = \Lambda(-0.0644 + 0.3 \times 1) = \Lambda(0.2356) = 0.5931, \text{ which is slightly larger than (a)}$$

$$c. DTIME = 3, \text{ Let } DTIME = x, \gamma_1 + \gamma_2 x = -0.2376 + 0.5311 \times 3 = 1.3557$$

$$\text{Logit model: } \frac{dP}{dx} = \frac{d\Lambda(\gamma_1 + \gamma_2 x)}{d(\gamma_1 + \gamma_2 x)} \cdot \frac{d(\gamma_1 + \gamma_2 x)}{dx} = \frac{e^{1.3557}}{1+e^{1.3557}} \cdot \frac{1}{1+e^{1.3557}} \times 0.5311 = 0.1629 \times 0.5311 = 0.0866$$

→ Increase probability 0.0866 with an increase of 10 minutes whose travel time is 30 minutes longer by bus

$$\gamma_1 = -0.0644, \gamma_2 = 0.3$$

$$\text{Linear prob. model: } \frac{dP}{dx} = \phi(-0.0644 + 0.3 \times 3) \cdot 0.3 = \phi(0.8356) \times 0.3 = \frac{1}{\sqrt{2\pi}} e^{-\frac{0.8356^2}{2}} \times 0.3 = 0.0844, \text{ slightly larger than Logit model}$$

→ Increase probability 0.0844 with an increase of 10 minutes whose travel time is 30 minutes longer by bus

$$d. DTIME = 5, \text{ Let } DTIME = x, \gamma_1 + \gamma_2 x = -0.2376 + 0.5311 \times 5 = 2.4179$$

$$\text{Logit model: } \frac{dP}{dx} = \frac{e^{2.4179}}{1+e^{2.4179}} \cdot \frac{1}{1+e^{2.4179}} \times 0.5311 = 0.0751 \times 0.5311 = 0.0399$$

→ Decrease probability 0.0399 with an decrease of 10 minutes whose travel time is 50 minutes longer by bus

$$\text{Linear prob. model: } \frac{dP}{dx} = \phi(-0.0644 + 0.3 \times 5) \times 0.3 = \frac{1}{\sqrt{2\pi}} e^{-\frac{1.9356^2}{2}} \times 0.3 = 0.0427, \text{ slightly larger than Logit model}$$

→ Decrease probability 0.0427 with an decrease of 10 minutes whose travel time is 50 minutes longer by bus

16.4 In Example 16.3, we illustrate the calculation of the likelihood function for the probit model in a small example. In this exercise, we will repeat that example using logit instead of probit. The logit model for the same example is  $P(y = 1) = \Lambda(\gamma_1 + \gamma_2 x)$ , where  $\Lambda(\bullet)$  is the logistic cdf in equation (16.7). The maximum likelihood estimates of the parameters are  $\hat{\gamma}_1 + \hat{\gamma}_2 x = -1.836 + 3.021x$ . The maximized value of the log-likelihood function is -1.612.

- Calculate the probability that  $y = 1$  if  $x = 1.5$ , given the values of the maximum likelihood estimates.
- Using the threshold 0.5 and the result in part (a), predict the value of  $y$  if  $x = 1.5$ , the first observation, given the values of the maximum likelihood estimates. Compare your prediction to the actual outcome  $y = 1$  in the first observation.
- Calculate the value of the likelihood function, illustrated in equation (16.14) but substituting equation (16.17) in place of  $\Phi(\bullet)$  and using the given  $N = 3$  data pairs, if the parameter values are  $\gamma_1 = -1$  and  $\gamma_2 = 2$ . Compare this value to the value of the likelihood function evaluated at the maximum likelihood estimates. Which is larger?
- For the logit model, the value of the likelihood function (16.14), with  $\Lambda(\bullet)$  in place of  $\Phi(\bullet)$ , will always be between zero and one. True or false? Explain.
- For the logit model, the value of the log-likelihood function (16.15), with  $\Lambda(\bullet)$  in place of  $\Phi(\bullet)$ , will always be negative. True or false? Explain.

| N | x   | y |
|---|-----|---|
| 1 | 1.5 | 1 |
| 2 | 0.6 | 1 |
| 3 | 0.7 | 0 |

$$a. P(y = 1 | x = 1.5) = \Lambda(-1.836 + 3.021 \times 1.5) = \Lambda(2.6955) = \frac{e^{2.6955}}{1+e^{2.6955}} = 0.9368$$

$$b. \because P(y = 1 | x = 1.5) = 0.9368 > 0.5 \therefore \hat{y} = 1 \text{ and actual outcome } y = 1 \rightarrow \text{correct model prediction}$$

$$c. \text{ Let } z_1 = -1 + 2 \times 1.5 = 2, z_2 = -1 + 2 \times 0.6 = 0.2, z_3 = -1 + 2 \times 0.7 = 0.4$$

$$\text{Likelihood function: } L = \prod_{i=1}^N [\Phi(z_i)]^{y_i} [1 - \Phi(z_i)]^{1-y_i} = \Phi(2) \Phi(0.2) (1 - \Phi(0.4)) = 0.195 < e^{-1.612} = 0.1995 \text{ (maximum likelihood)}$$

d. Since  $\Phi(\cdot)$  is cdf of standard normal distribution,  $0 \leq \Phi(x) \leq 1 \quad \forall x \in \mathbb{R}$

$$\Rightarrow 0 \leq \Phi(x), 1 - \Phi(x) \leq 1 \therefore L = \prod_{i=1}^N (1 - \Phi(z_i))^{1-y_i} \Phi(z_i)^{y_i} \Rightarrow 0 \leq L \leq 1 \Rightarrow \text{The statement is true}$$

e. By (d)  $0 \leq L \leq 1 \Rightarrow -\infty < \ln L \leq 0 \Rightarrow \text{If } \ln L = 0 \Rightarrow \Phi(z_i) = 0 \text{ or } 1 \Rightarrow z_i = \pm \infty$

$\Rightarrow \ln L$  is impossible to be 0  $\Rightarrow \ln L < 0 \Rightarrow \text{The statement is true}$

**16.18** Mortgage lenders are interested in determining borrower and loan characteristics that may lead to delinquency or foreclosure. In the data file *lasvegas* are 1000 observations on mortgages for single family homes in Las Vegas, Nevada during 2008. The variable of interest is *DELINQUENT*, an indicator variable = 1 if the borrower missed at least three payments (90 + days late), but 0 otherwise. Explanatory variables are *LVR* = the ratio of the loan amount to the value of the property; *REF* = 1 if purpose of the loan was a “refinance” and = 0 if loan was for a purchase; *INSUR* = 1 if mortgage carries mortgage insurance, 0 otherwise; *RATE* = initial interest rate of the mortgage;

*AMOUNT* = dollar value of mortgage (in \$100,000); *CREDIT* = credit score, *TERM* = number of years between disbursement of the loan and the date it is expected to be fully repaid, *ARM* = 1 if mortgage has an adjustable rate, and = 0 if mortgage has a fixed rate.

- Estimate the linear probability (regression) model explaining *DELINQUENT* as a function of the remaining variables. Use White heteroskedasticity robust standard errors. Are the signs of the estimated coefficients reasonable?
- Use logit to estimate the model in (a). Are the signs and significance of the estimated coefficients the same as for the linear probability model?
- Compute the predicted value of *DELINQUENT* for the 500th and 1000th observations using both the linear probability model and the logit model. Interpret the values.
- Construct a histogram of *CREDIT*. Using both linear probability and logit models, calculate the probability of delinquency for *CREDIT* = 500, 600, and 700 for a loan of \$250,000 (*AMOUNT* = 2.5). For the other variables, let the loan to value ratio (*LVR*) be 80%, the initial interest rate is 8%, all indicator variables take the value 0, and *TERM* = 30. Discuss similarities and differences among the predicted probabilities from the two models.
- Using both linear probability and logit models, compute the marginal effect of *CREDIT* on the probability of delinquency for *CREDIT* = 500, 600, and 700, given that the other explanatory variables take the values in (d). Discuss the interpretation of the marginal effect.
- Construct a histogram of *LVR*. Using both linear probability and logit models, calculate the probability of delinquency for *LVR* = 20 and *LVR* = 80, with *CREDIT* = 600 and other variables set as they are in (d). Compare and contrast the results.
- Compare the percentage of correct predictions from the linear probability model and the logit model using a predicted probability of 0.5 as the threshold.
- As a loan officer, you wish to provide loans to customers who repay on schedule and are not delinquent. Suppose you have available to you the first 500 observations in the data on which to base your loan decision on the second 500 applications (501–1,000). Is using the logit model with a threshold of 0.5 for the predicted probability the best decision rule for deciding on loan applications? If not, what is a better rule?

a.

Coefficients:

|             | Estimate   | Std. Error | t value | Pr(> t ) |     |
|-------------|------------|------------|---------|----------|-----|
| (Intercept) | 0.6884913  | 0.2112456  | 3.259   | 0.001155 | **  |
| lvr         | 0.0016239  | 0.0007846  | 2.070   | 0.038732 | *   |
| ref         | -0.0593237 | 0.0238299  | -2.489  | 0.012957 | *   |
| insur       | -0.4815849 | 0.0236365  | -20.375 | < 2e-16  | *** |
| rate        | 0.0343761  | 0.0085999  | 3.997   | 6.88e-05 | *** |
| amount      | 0.0237680  | 0.0126699  | 1.876   | 0.060957 | .   |
| credit      | -0.0004419 | 0.0002018  | -2.190  | 0.028782 | *   |
| term        | -0.0126195 | 0.0035386  | -3.566  | 0.000379 | *** |
| arm         | 0.1283239  | 0.0318867  | 4.024   | 6.15e-05 | *** |

The results indicate that **lvr**, **rate**, and **arm** have positive and statistically significant effects on the probability of delinquency. In particular, a higher loan-to-value ratio increases the likelihood of mortgage default, which is consistent with the idea that borrowers with less home equity face greater financial risk. The positive coefficient on **arm** suggests that adjustable-rate mortgages are associated with a higher probability of delinquency.

On the other hand, **ref**, **insur**, **credit**, and **term** have negative and statistically

significant coefficients. The negative coefficient on **credit** implies that borrowers with better credit scores are less likely to become delinquent. Mortgage insurance also significantly reduces the probability of delinquency. After using robust standard errors, **amount** is no longer statistically significant at the 5% level.

Overall, the signs of the estimated coefficients are economically reasonable and generally consistent with expectations.

b.

Coefficients:

|             | Estimate  | Std. Error | z value | Pr(> z ) |     |
|-------------|-----------|------------|---------|----------|-----|
| (Intercept) | 1.665303  | 1.992157   | 0.836   | 0.403195 |     |
| lvr         | 0.013519  | 0.008199   | 1.649   | 0.099164 | .   |
| ref         | -0.525064 | 0.230491   | -2.278  | 0.022725 | *   |
| insur       | -3.122105 | 0.216906   | -14.394 | < 2e-16  | *** |
| rate        | 0.308656  | 0.082771   | 3.729   | 0.000192 | *** |
| amount      | 0.217606  | 0.113769   | 1.913   | 0.055786 | .   |
| credit      | -0.003564 | 0.001933   | -1.844  | 0.065232 | .   |
| term        | -0.133037 | 0.034597   | -3.845  | 0.000120 | *** |
| arm         | 1.405649  | 0.371355   | 3.785   | 0.000154 | *** |

The estimated coefficients exhibit the same signs as those obtained from the LPM.

Variables such as **lvr**, **rate**, **amount**, and **arm** have positive coefficients, while **ref**, **insur**, **credit**, and **term** have negative coefficients. This consistency suggests that the relationship between borrower characteristics and delinquency is robust across model specifications.

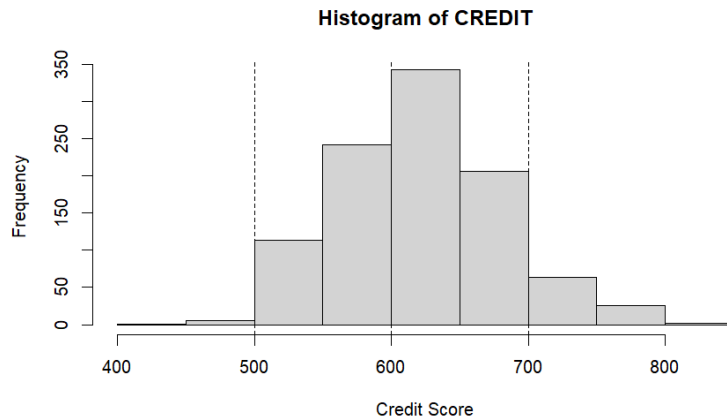
Overall, the logit model confirms the main findings from the LPM and provides a more appropriate framework for modeling binary outcomes.

c. For observation 500, the predicted probability of delinquency is 0.1828 under the LPM and 0.1336 under the logit model. Both models suggest a relatively low probability of delinquency for this borrower.

For observation 1000, the predicted probability is 0.5785 under the LPM and 0.6266 under the logit model. In this case, both models predict a relatively high likelihood of delinquency.

The two models lead to similar conclusions regarding the relative risk levels of the two borrowers. Observation 1000 appears substantially riskier than observation 500.

d.



For credit score is 500, 600, 700. Under the LPM, the predicted probabilities are 0.5533, 0.5091, and 0.4649, respectively. Under the logit model, the corresponding probabilities are 0.4967, 0.4086, and 0.3261.

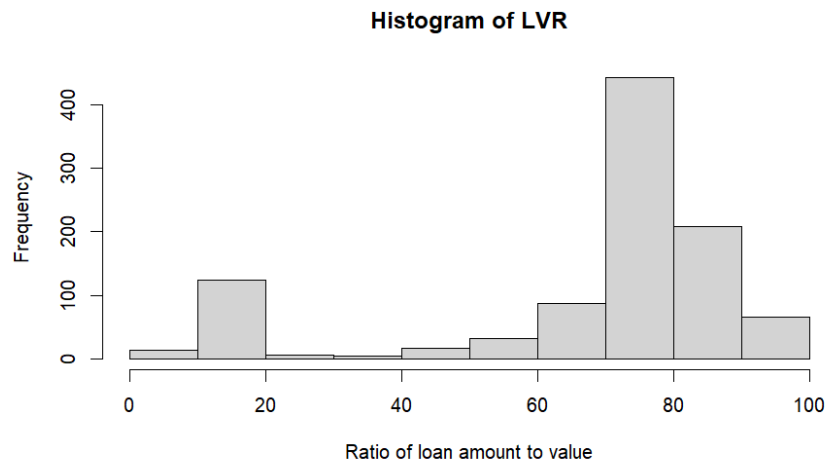
The results show a clear negative relationship between credit score and delinquency risk. As the credit score increases from 500 to 700, the predicted probability of delinquency declines substantially in both models. The decline is particularly pronounced in the logit model, reflecting the nonlinear nature of the logistic function.

e. For the LPM, the marginal effect of credit score is constant and equal to  $-0.000442$ . This means that a one-point increase in the credit score reduces the probability of delinquency by approximately 0.044 percentage points, regardless of the borrower's characteristics.

For the logit model, the marginal effects vary across borrowers. The estimated marginal effects are  $-0.000891$ ,  $-0.000861$ , and  $-0.000783$  for credit scores of 500, 600, and 700, respectively.

The results indicate that improvements in credit quality reduce delinquency risk. Moreover, the magnitude of the marginal effect becomes slightly smaller as the credit score increases. This pattern reflects the nonlinear structure of the logit model, where marginal effects depend on the underlying probability level.

f.



Under the LPM, the predicted probabilities are 0.4117 and 0.5091, respectively. Under the logit model, the probabilities are 0.2349 and 0.4086. The results indicate that a higher loan-to-value ratio substantially increases the probability of delinquency. Borrowers with higher leverage have less equity in their homes and therefore face greater default risk. This finding is consistent with economic intuition and the estimated positive coefficient on LVR.

g.

| Predicted |     |     |
|-----------|-----|-----|
| True      | 0   | 1   |
| 0         | 727 | 74  |
| 1         | 68  | 131 |

Confusion matrices for Linear probability

| Predicted |     |     |
|-----------|-----|-----|
| True      | 0   | 1   |
| 0         | 736 | 65  |
| 1         | 81  | 118 |

Confusion matrices for Logit Model

Using a classification threshold of 0.5, the prediction accuracy of the LPM is 85.8%, while the logit model achieves an accuracy of 85.4%.

The two models perform similarly in terms of overall classification accuracy. The confusion matrices show that the logit model produces fewer false positives but more false negatives than the LPM. Therefore, although the overall accuracy is similar, the types of classification errors differ between the two models.

Overall, both models provide reasonably strong predictive performance, with no substantial advantage for either model in terms of raw accuracy.

h. The first 500 observations were used as the training sample, and the remaining 500

observations were used as the testing sample. Using a classification threshold of 0.5, the out-of-sample prediction accuracy is 88%.

To determine whether 0.5 is the optimal threshold, several alternative cutoff values were examined. The results show that the highest accuracy, 88.4%, occurs when the threshold is approximately 0.489 or 0.490. This accuracy is slightly higher than the 88.0% achieved with the conventional threshold of 0.5. Therefore, if the objective is to maximize classification accuracy, a threshold slightly below 0.5 appears to be preferable.